

Project: Green Hydrogen Supply Chain

Population growth, higher living standards, and industrial development have intensified climate change, making continued dependence on fossil fuels unsustainable. To meet international climate goals such as the Paris Agreement and the EU “Fit for 55” package, green hydrogen (H_2) emerges as key decarbonization solution, especially for hard-to-abate sectors. In 2023, global H_2 demand reached 97 Mt, mainly in refining and chemicals, while low-emission H_2 accounted for less than 1 Mt (International Energy Agency, 2024).

Scaling up green H_2 faces major challenges, including cost reduction, advancing technology maturity, and expanding infrastructure, with the ultimate goal of making low-carbon H_2 competitive across industries. Among the EU member states, Portugal stands out for its abundant renewable resources and has adopted a National Strategy for Hydrogen (EN- H_2). By 2030, the country aims for green H_2 to supply from 2% to 5% of industrial energy consumption, strengthening its ambition to become a strategic hub in the European H_2 economy (Presidência do Conselho de Ministros, 2020). Advancing this transition requires the establishment and comprehensive characterization of the green H_2 supply chain (GHSC).

Green Hydrogen Production Technologies, Storage and Transportation modes

H_2 is classified as “green” when produced through processes with no net carbon dioxide (CO_2) emissions. Currently, the most mature and deployed technology for green H_2 production is water electrolysis, powered by renewable electricity, which splits water into H_2 and oxygen (O_2). Carbon-neutral processes, such as biomass gasification, are also considered, with biomass being a natural option as a renewable source to feed H_2 production processes. The produced H_2 can be stored either as a compressed gas, in pressure vessels, or as a cryogenic liquid, in spherical tanks. Its physical form determines the mode of transportation: gaseous H_2 is typically delivered via tube trailers or pipelines, while liquid H_2 is transported in cryogenic tanker trucks.

Project Goal

The objective of this project is to design an optimized GHSC in Portugal, with a particular focus on meeting H_2 demand for the country’s industrial sector. The goal is to develop a robust and efficient supply chain capable of addressing this demand, while tackling key challenges such as high infrastructure costs and evolving market penetration.

Problem Description

The GHSC under consideration is structured into four main stages: (i) primary energy sources and feedstock; (ii) production; (iii) storage; and (iv) distribution. An overview of the GHSC structure is presented in Figure 1. For simplification purposes, consider the following assumptions:

- (i) **Primary energy sources and feedstock:** Only renewable energy sources (RES) are considered, specifically solar and wind. Water, the main feedstock for electrolysis, is assumed to be fully available, therefore, water availability and consumption is not taken into account.
- (ii) **Production technology:** Water electrolysis is the only production technology to be considered for the generation of H_2 . Among the various electrolysis technologies, Proton Exchange Membrane (PEM) electrolysis is selected due to its suitability for coupling with RES. Three plant sizes are available: small, medium or large, each operating at a fixed efficiency of 75%.
- (iii) **Storage:** H_2 is assumed to be stored in gaseous form, with different storage capacities.
- (iv) **Transport:** H_2 is distributed exclusively by tube trailers (on-road transportation), in its gaseous form.

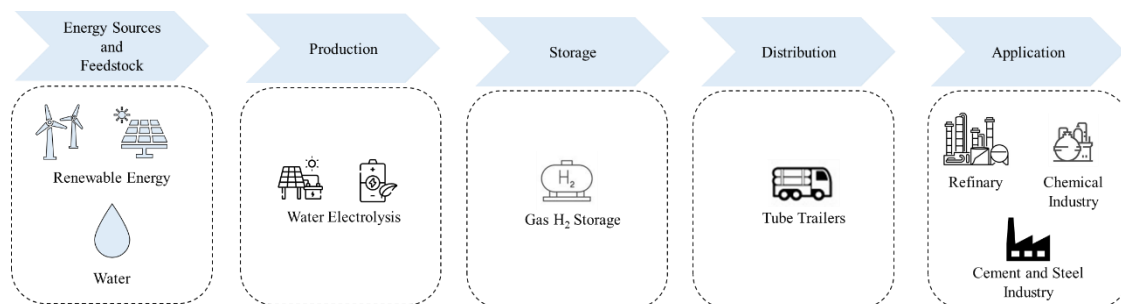


Figure 1. Overview of the GHSC structure considered for this work

The H_2 demand of the Portuguese industrial sector, as well as the location and availability of RES, is provided by district in Table 1 (data was obtained from Direção-Geral de Energia e Geologia (DGEG) and Energias Endógenas de Portugal (E2P)). **For conversion purposes, assume that the production of 1 tonne of H_2 requires 0,051 GWh of electricity.**

Table 1. H₂ demand and RES availability, at a yearly basis, per district (DGEG (2023), E2P (2024))

Location	District	H ₂ demand (tonnes/year)	RES Availability	
			Wind (GWh/year)	Solar (GWh/year)
L1	Aveiro	220 250	290	39
L2	Beja	-	474	1 268
L3	Braga	11 560	1 099	25
L4	Bragança	6	536	266
L5	Castelo Branco	250	4237	9
L6	Coimbra	49 934	4232	182
L7	Évora	996	-	693
L8	Faro	7	1 511	2 287
L9	Guarda	58	3 547	50
L10	Leiria	184 585	1 992	-
L11	Lisboa	51 645	2 364	1 221
L12	Portalegre	56	52	130
L13	Porto	62 330	531	49
L14	Santarém	30 687	1 075	418
L15	Setúbal	457 668	119	1 148
L16	Viana do Castelo	2 488	2 737	-
L17	Vila Real	184	4 406	1
L18	Viseu	3 896	7 709	7

As previously contextualized, in practice, the H₂ demand (Table 1) cannot be fully met in the short term, since large-scale deployment of the supply chain is still under development and international targets extend to 2050. To support this transition, the EN-H2 strategy has defined interim targets and indicative trajectories. These targets, shown in Table 2, establish the minimum percentage of H₂ to be incorporated into the industrial sector's energy consumption by 2030, 2040, and 2050.

Table 2. Targets for the incorporation of H₂ in the industrial sector (Presidência do Conselho de Ministros, 2020.)

Year	Target
2030	2% - 5%
2040	10% - 15%
2050	20% - 25%

Technical and Economic Information

- H₂ Production Plants and PEM Electrolyser

A summary of the technical and economic parameters of H₂ production plants is presented in Table 3. Three different sizes of H₂ production plants are considered: small (S), medium (M), and large (L). Consider the operational efficiency of the electrolyser for the production process.

Table 3. Technical and economic parameters of H₂ production facilities

Production technology (size)	Capital Cost (k€)	Capacity (tones H₂/day)	PEM Electrolyser Efficiency (%)
Water Electrolysis PEM (S)	15 500	500	75
Water Electrolysis PEM (M)	231 800	1 000	75
Water Electrolysis PEM (L)	741 500	2 000	75

- Storage Facilities

The technical and economic parameters of H₂ storage facilities are provided in Table 4. Five different storage capacities are available: tank 1, tank 2, tank 3, tank 4, and tank 5.

Table 4. Technical and economic parameters of H₂ storage facilities

Storage Facility (size)	Capital Cost (k€)	Capacity (tons H₂/day)
Tank 1	25 000	1 000
Tank 2	50 000	2 000
Tank 3	125 000	2 500
Tank 4	250 000	3 000
Tank 5	500 000	3 500

- Transportation

Table 5. Transportation mode costs (gaseous H₂)

Parameter	Tube Trailer Cost
Shipping cost (€/tonne H ₂ .km)	0.35 to 0.50

- Distance between districts in kilometres (km)

Table 6. Distance by road between districts, in km.

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
1	0	200	118	324	274	217	178	273	542	444	510	210	383	751	272	545	363	462
2	200	0	97	94	200	176	76	170	428	225	291	127	195	528	86	327	260	360
3	118	97	0	214	164	107	171	266	536	334	400	100	273	637	162	435	355	493
4	324	94	214	0	195	171	155	138	343	141	206	122	79	443	63	242	172	300
5	274	200	164	195	0	62	273	331	521	320	385	74	258	622	148	421	365	478
6	217	176	107	171	62	0	251	309	499	298	363	57	236	600	126	399	344	457
7	178	76	171	155	273	251	0	99	369	253	318	200	262	555	159	354	189	289
8	273	170	266	138	331	309	99	0	275	159	224	257	169	461	199	260	95	195
9	542	428	536	343	521	499	369	275	0	205	176	448	283	147	389	142	181	80
10	444	225	334	141	320	298	253	159	205	0	80	245	81	307	186	117	162	163
11	510	291	400	206	385	363	318	224	176	80	0	314	150	281	255	53	230	137
12	210	127	100	122	74	57	200	257	448	245	314	0	187	551	76	349	294	407
13	383	195	273	79	258	236	262	169	283	81	150	187	0	383	119	180	172	239
14	751	528	637	443	622	600	555	461	147	307	281	551	383	0	491	244	382	226
15	272	86	162	63	148	126	159	199	389	186	255	76	119	491	0	289	234	347
16	545	327	435	242	421	399	354	260	142	117	53	349	180	244	289	0	193	100
17	363	260	355	172	365	344	189	95	181	162	230	294	172	382	234	193	0	103
18	462	360	493	300	478	457	289	195	80	163	137	407	239	226	347	100	103	0

Based on the information provided, answer the following questions in a thorough and critical manner.

Note: All assumptions and additional data that you consider necessary must be clearly identified and justified.

1. Formulate a mathematical model to optimize the GHSC network design and planning, while minimizing the total supply chain costs. This should be accomplished by determining the number, location, and size of production plants and storage facilities, as well as the allocation of resources and product flows. Regarding distribution, the number of tube trailers required for transportation from production plants to storage facilities, and from storage facilities to consumers, should be displayed. Use a solver of your choice to implement and solve the model.

2. Based on the developed model, perform a sensitivity analysis on the most relevant parameters. From the identification of the critical parameters, discuss how an environmental assessment could be conducted. Indicate what kind of data would be required and emphasize how this assessment could inform real-world decisions regarding environmental sustainability and address challenges in H₂ supply chains.

3. Focusing on the production stage of the GHSC, consider H₂ generation from two different renewable sources: water and biomass. Establish a comparative analysis between these two production routes, identifying their respective advantages and disadvantages. In your discussion, consider factors such as scalability and deployment feasibility, including the potential for centralized versus decentralized supply chain configurations.

Report guidelines – video report

1. **Prepare a presentation using PowerPoint** with no more than 11 slides (excluding the cover slide).

The structure must be as follows:

- Cover slide: name, number, and photo of the group members
- Short introduction about the topic and the main questions (1 slide)
- Present the strategy used to obtain your results, as well as summary tables/charts with the main results; discuss the results and relate them with the theoretical concepts - this is the most important content of your work (maximum 9 slides)
- Conclusions – include the main limitations of the model and possible improvements (as future work recommendations) (1 slide)

2. **Record a video of the PowerPoint presentation with narration.**

All group members must participate in the video narration. The video's maximum duration is 15 minutes. For a tutorial on how to make a video in PowerPoint you can check the following link: <https://www.youtube.com/watch?v=D8jV3w4TOVw>

3. **Gather all files used to reach the results** presented and discussed in the video report (e.g., excel files, gams, cplex, python).
4. **Create a zip file** that includes all the files and the video. The zip file name must be “S#1_G#2”, where #1 corresponds to the Shift number and #2 corresponds to the Group number.
5. **Submit the zip file** in Fenix by the 15th of October.